

BRAC UNIVERSITY
Department of Computer Science and Engineering
CSE 260: Digital Logic Design

Examination: Quiz 1
Duration: 25 Minutes

Semester: Summer 2025
Full Marks: 15

Answer the following questions. You **MUST** show your workings/calculations where applicable.
Figures in the right margin indicate marks.

Name: <u>Solution</u>	Section:	ID:
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1. Perform the following conversion: $(100101010)_2 = (?)_{17}$
[CO1] **Note:** You **must** show all the necessary conversions. [6]

2. A secure keypad entry system uses Excess-5 code to encode individual decimal digits for internal processing. During a routine check, a technician inspects a 16-bit value received from the system: $(1010001111000110)_{\text{Ex-5}}$
[CO1] As a digital logic designer:
 - a) Determine whether this 16-bit sequence is a valid encoding of a 4-digit decimal number using Excess-5 or not. [1]
 - b) If you detect any invalid group(s), provide a logical explanation for why they are invalid based on the structure and rules of Excess-5 encoding. [2]

3. Divide $(1112311)_4$ by $(53)_6$. Find the quotient and remainder. Perform the division in base-4 and make sure your final answers are in base-4.
[CO1] **Note:** You **must** show the necessary calculations. [6]

① $(100101010)_2 \rightarrow$ negative number

011010101
 $+ 1$

 $(011010110)_2 \rightarrow (-214)_{10}$

$17 \overline{) 214}$
 $17 \overline{) 12 - (10) \rightarrow A}$
 $0 - (12) \rightarrow C$

$\therefore (100101010)_2 = (-CA)_{17}$

2

$$\begin{array}{r} (1010 \\ 10 \end{array} \quad \begin{array}{r} 0011 \\ 3 \end{array} \quad \begin{array}{r} 1100 \\ 12 \end{array} \quad \begin{array}{r} 0110 \\ 6 \end{array} \text{ Ex-5}$$

↓

this is the error group.

$$\begin{array}{l} 0 \rightarrow 0+5 = 5 \rightarrow 0101 \\ \vdots \\ 9 \rightarrow 9+5 = 14 \rightarrow 1110 \end{array} \left. \vphantom{\begin{array}{l} 0 \\ \vdots \\ 9 \end{array}} \right\} \text{ valid groups.}$$

3

$$(1112311)_4 / (53)_6$$

$$(1112311)_4 / (201)_4$$

$$\begin{array}{r} \overline{11111} \\ 201 \overline{) 1112311} \\ \underline{0} \\ \overline{11} \\ \underline{0} \\ \overline{111} \\ \underline{0} \\ \overline{1112} \\ \underline{1002} \\ \overline{1103} \\ \underline{1002} \\ \overline{1011} \\ \underline{1002} \\ \overline{31} \\ \underline{-0} \\ \overline{31} \end{array}$$

$$(53)_6 \rightarrow (201)_4$$

$$201 \times 0 = 0$$

$$201 \times 1 = 201$$

$$201 \times 2 = 1002$$

$$201 \times 3 = 1203$$

$$\text{quotient} = (2220)_4$$

$$\text{rem.} = (31)_4$$